

SHUNGNAC, AK, USA

WMO: 701719

Lat: 66.888N

Lon: 157.162W

Elev: 206

StdP: 14.59

Time Zone: -9.00 (NAK)

Period: 86-19

WBAN: 00490

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.8	(c) -40.6	(d) -45.7	(e) 0.4	(f) -35.5	(g) -38.5	(h) 0.6	(i) -26.2	(j) 34.7	(k) 18.9	(l) 32.2	(m) 20.8	(n) 5.1	(o) 110	(p) 0.775

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.9	(c) 78.7	(d) 59.5	(e) 74.6	(f) 58.2	(g) 71.3	(h) 56.9	(i) 62.1	(j) 74.3	(k) 60.1	(l) 70.8	(m) 58.5	(n) 68.0	(o) 7.5	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.3	70.7	64.1	55.6	66.4	62.3	54.3	63.2	60.9	27.8	73.9	26.5	70.7	25.4	67.9	68.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.5	25.6	22.4	-44.2	83.3	7.7	4.5	-49.8	86.6	-54.3	89.2	-58.6	91.7	-64.1	95.0
			-35.9	63.9	7.6	1.9	-41.3	65.3	-45.8	66.4	-50.1	67.4	-55.6	68.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	-4.3	2.9	3.4	22.4	41.6	56.7	59.8	53.9	43.7	27.4	5.0	-1.6	
	DBStd	27.15	23.15	17.49	13.95	14.58	10.70	6.76	6.07	5.86	7.46	10.66	15.62	18.13	
	HDD50	9486	1684	1320	1445	829	293	21	3	27	210	702	1351	1601	
	HDD65	14259	2149	1740	1910	1278	725	259	182	344	638	1167	1801	2066	
	CDD50	732	0	0	0	1	33	220	308	149	21	0	0	0	
	CDD65	32	0	0	0	0	0	9	22	1	0	0	0	0	
	CDH74	423	0	0	0	0	2	130	284	7	0	0	0	0	
	CDH80	92	0	0	0	0	0	18	74	0	0	0	0	0	
Wind	WSAvg	8.5	10.2	10.5	8.0	8.8	7.8	7.0	6.1	6.7	7.7	9.8	9.2	10.8	
Precipitation	PrecAvg	16.7	1.1	1.1	0.8	0.6	0.7	1.4	2.2	2.6	2.0	1.6	1.3	1.2	
	PrecMax	21.5	2.5	3.0	2.5	2.1	1.3	3.9	3.9	6.1	4.2	4.5	3.3	2.8	
	PrecMin	12.6	0.1	0.1	0.1	0.1	0.2	0.5	1.3	1.0	0.9	0.6	0.1	0.4	
	PrecStd	2.7	0.6	0.7	0.5	0.4	0.4	0.6	0.7	1.0	0.9	0.8	0.9	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	37.4	39.4	56.3	73.2	82.7	87.0	74.4	65.9	51.0	38.0	34.5	
		MCWB	36.1	30.7	31.3	43.1	54.4	60.6	63.9	59.8	52.8	40.0	33.0	31.5	
	2%	DB	34.4	33.9	34.1	50.9	68.4	77.8	81.2	70.9	61.4	45.8	32.1	32.0	
		MCWB	31.4	30.3	28.0	39.7	51.5	58.2	60.8	57.4	50.0	38.4	29.4	29.2	
	5%	DB	29.4	30.7	29.6	46.0	63.0	74.2	76.5	67.5	57.7	43.5	28.4	28.0	
		MCWB	26.7	28.0	25.8	37.2	49.3	56.6	59.8	56.3	48.6	38.0	26.3	26.0	
	10%	DB	25.2	27.5	24.8	41.8	58.4	70.6	72.3	64.2	54.6	40.9	25.0	23.3	
		MCWB	23.0	24.6	21.6	34.9	47.1	55.2	58.9	55.0	48.0	36.4	22.9	21.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	33.2	32.1	43.8	56.0	62.3	65.2	61.8	54.5	43.2	33.1	32.5
			MCDB	40.7	34.7	38.7	55.8	71.2	75.2	80.3	71.1	61.8	45.7	37.4	33.4
2%		WB	32.6	30.6	28.8	39.8	52.9	60.0	63.3	59.4	52.4	40.8	30.0	29.7	
		MCDB	35.2	33.7	33.5	49.6	66.0	73.3	76.8	66.7	57.9	44.1	32.0	32.1	
5%		WB	28.0	28.1	25.7	37.4	50.5	58.4	61.6	57.6	50.5	38.6	26.6	26.4	
		MCDB	29.8	31.1	30.0	45.3	59.9	70.6	73.6	64.9	55.5	41.8	28.3	28.8	
10%		WB	24.1	24.7	21.9	35.2	48.3	56.8	59.7	56.2	49.0	36.9	23.2	22.0	
		MCDB	26.6	27.4	24.7	41.4	57.2	67.8	69.7	62.2	54.0	40.2	25.0	23.7	
Mean Daily Temperature Range		5% DB	MDBR	12.2	14.5	18.9	17.5	19.0	18.7	15.9	14.0	13.6	10.7	11.2	12.5
			MCDBR	8.9	11.3	19.3	17.6	26.5	24.4	25.2	18.7	18.9	10.1	10.0	10.1
	MCWBR		7.4	9.4	15.3	10.8	13.2	9.7	10.2	8.9	10.3	6.3	8.6	9.3	
	5% WB	MCDBR	8.9	11.5	18.6	16.0	22.7	20.8	21.6	15.1	15.3	8.8	9.7	10.3	
		MCWBR	7.6	9.7	14.8	10.0	11.4	9.0	9.3	7.8	8.9	6.4	8.6	9.6	
Clear-Sky Solar Irradiance	taub	0.173	0.241	0.260	0.322	0.355	0.364	0.367	0.353	0.306	0.257	0.169	N/A		
	taud	1.913	2.127	2.238	2.132	2.253	2.362	2.378	2.437	2.514	2.282	1.939	N/A		
	Ebn at Noon	67	193	256	262	266	267	261	253	242	186	64	0		
	Edh at Noon	4	16	24	36	36	33	32	27	19	13	4	0		
All-Sky Solar Radiation	RadAvg	25	208	782	1485	1729	1809	1451	1084	659	246	50	0		
	RadStd	3	19	54	77	171	160	119	135	85	36	6	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page